CO-1 AT-1

Technical Presentation

1. NFA to DFA conversion

$\Sigma = \{0, 1\}$

States =

$Q = \{A, B, C\}$

Start State: A

Final State: C

State	input	output
A	$\{A\}$	$\{B\}$
B	$\{A, C\}$	$\emptyset$
C	$\emptyset$	$\{A, B\}$

Step 1:

Start State : $\{A\}$

for input:

$(\{A\}, 0) = \{A\}$

for input : 1

$(\{A\}, 1) = \{B\}$

So DFA

$\{A\}, \{B\}$

Step: 2

For Input:

$\{B\}, \{0\} = \{A, C\}$

For input : 1

$(\{B\}, 1) = \emptyset$

new state :

$\{A, C\}, \emptyset$

Step 3

process $\{A, C\}$,

$A \rightarrow A$

$C \rightarrow \emptyset$

Therefore

$(\{A, C\}, 0) = \{A\}$

For input 1 :

$A \rightarrow B$

$C \rightarrow \{A, B\}$

Therefore :

$(\{A, C\}, 1) = \{A, B\}$

New State :

$\{A, B\}$

AC is a final state.

Step 4:

Process {A, B}

For input 0:

A → A

B → {A, C}

Therefore:

({A, B}, 0) = {A, C}

Step 5:

Process ∅

For both inputs

(∅, 0) = ∅

(∅, 1) = ∅

So, ∅ is the dead state

Final transition table

DFA State	0	1	Final
{A}	{A}	{B}	No
{B}	{A, C}	∅	No
* {A, C}	{A}	{A, B}	Yes
{A, B}	{A, C}	{B}	No
∅	∅	∅	No

2 Mathematical Introduction in automata theory

Mathematical Introduction is a proof technique used to prove that a statement is true for all positive integers.

- Strings of length n
- Languages
- Number of symbols in a String
- properties of automata after reading in symbols

Step 1:

Best case

First, prove that the statement is true for the smallest value

$n = 1$

we, prove => P(1) is true

Step 2

Inductive hypothesis.

P(k) is true

Assume that the statement is true for some (+) ve integer k. This is assumption is called Inductive hypothesis.

Step 3:

Inductive step

P(k+1)

P(k) = P(k+1)

Using the assumptions that P(k) is true prove that P(k+1)

1) P(1) is true
2) P(k) = P(k+1)

So,

P(n) is true for all positive integers

Assume :-

$$1^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

for : $k = 1$

$$\Rightarrow \frac{k(k+1)(2k+1)}{2} + (k+1)^2$$

$$\Rightarrow \frac{(k+1)(k+2)(2k+3)}{6}$$

So,

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ☀ | ?

States | Alphabet | Transitions

FROM
C

SYMBOL
1

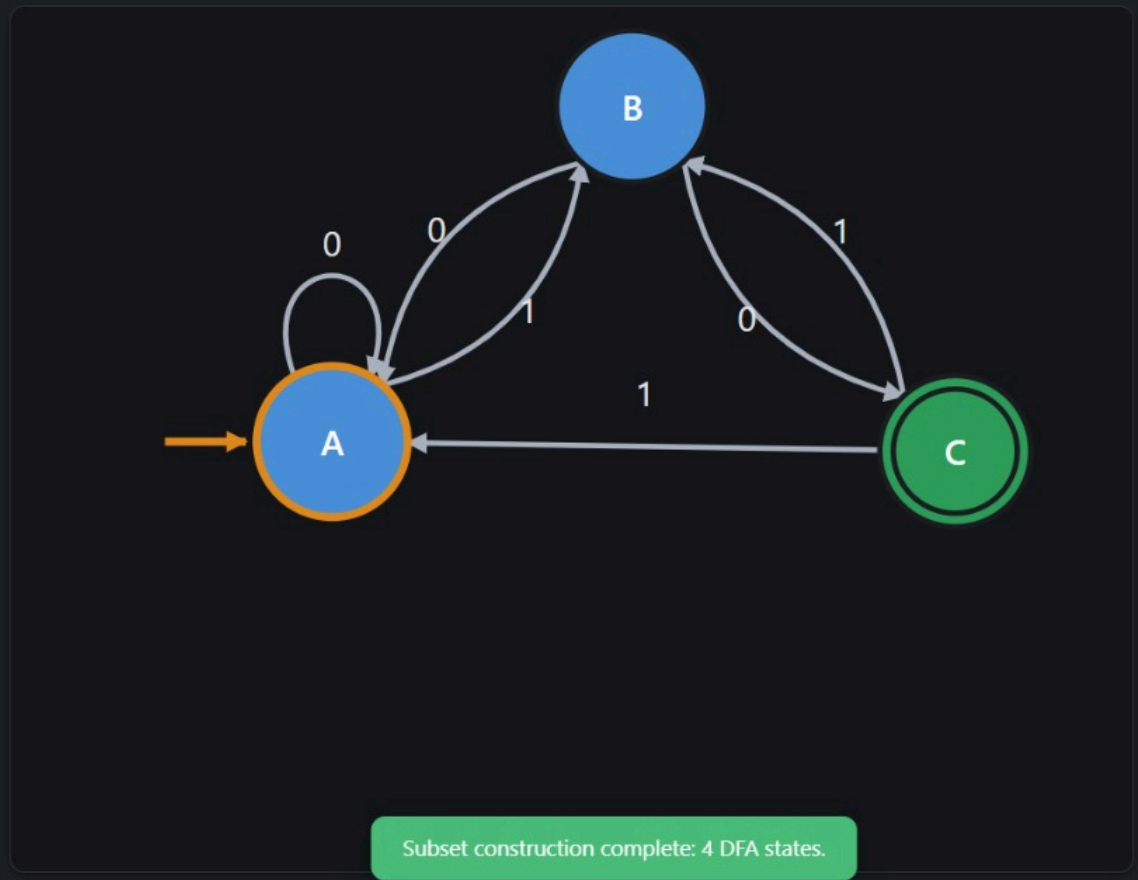
TO
B

Add transition

A —0→ A
A —1→ B
B —0→ A
B —0→ C
C —1→ A
C —1→ B

Delete transition

DFA: binary strings ending in 1 | NFA



Subset construction complete: 4 DFA states.

State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

NFA → DFA (subset construction)

Minimize DFA

DFA → NFA

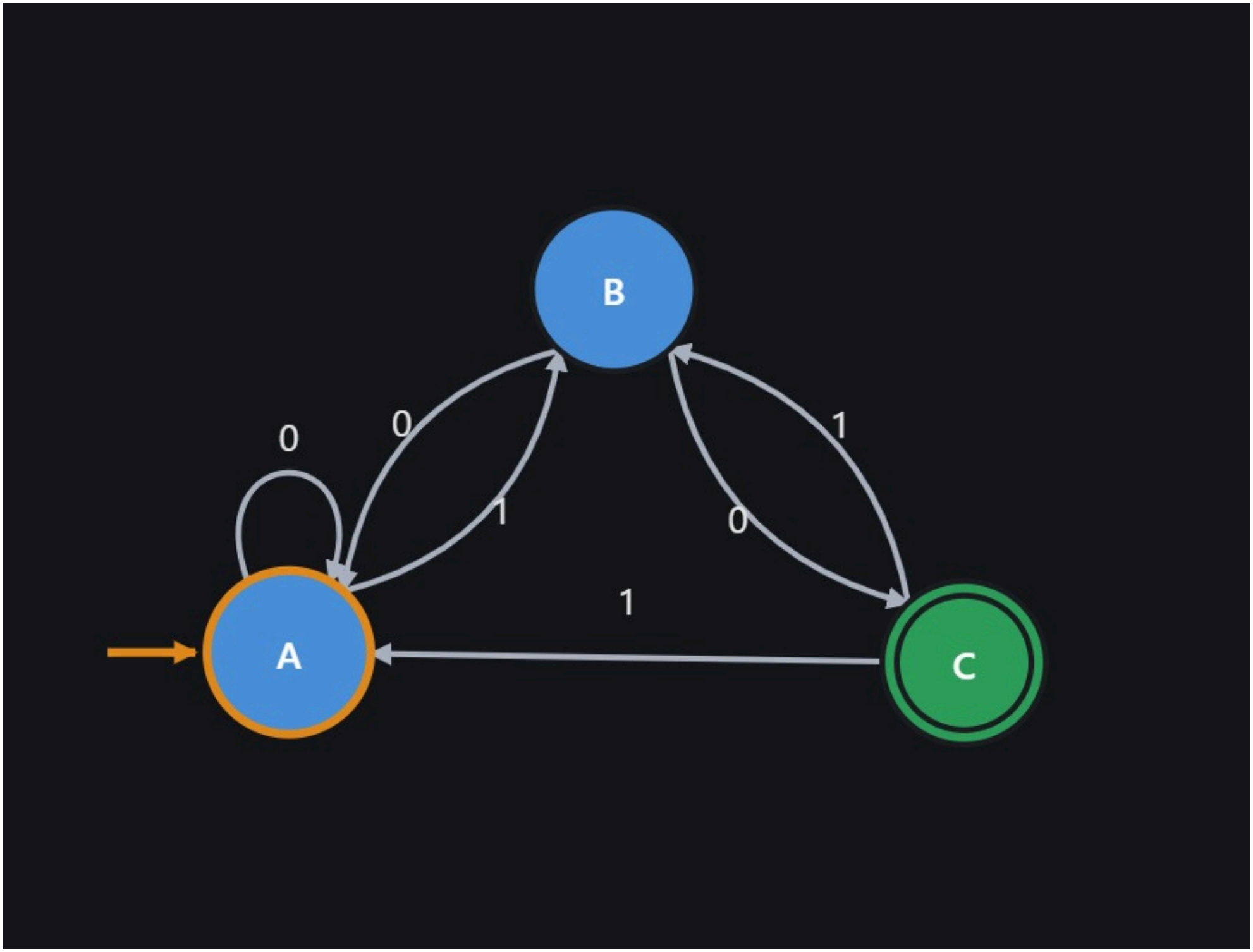
SUBSET CONSTRUCTION TABLE

DFA state	NFA states	0	1
→ {A}	{A}	{A}	{B}
{B}	{B}	{A,C}	∅
* {A,C}	{A, C}	{A}	{A,B}
{A,B}	{A, B}	{A,C}	{B}

→ start state · * accepting state · 4 DFA states from 3 NFA states.

Use as current automaton

INPUT STRING e.g. 1011 | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭ | Enter a string and press Simulate.


B
0
0
1
1
0
1
A
C

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ☀ | ?

States | Alphabet | Transitions

FROM
A

SYMBOL
0

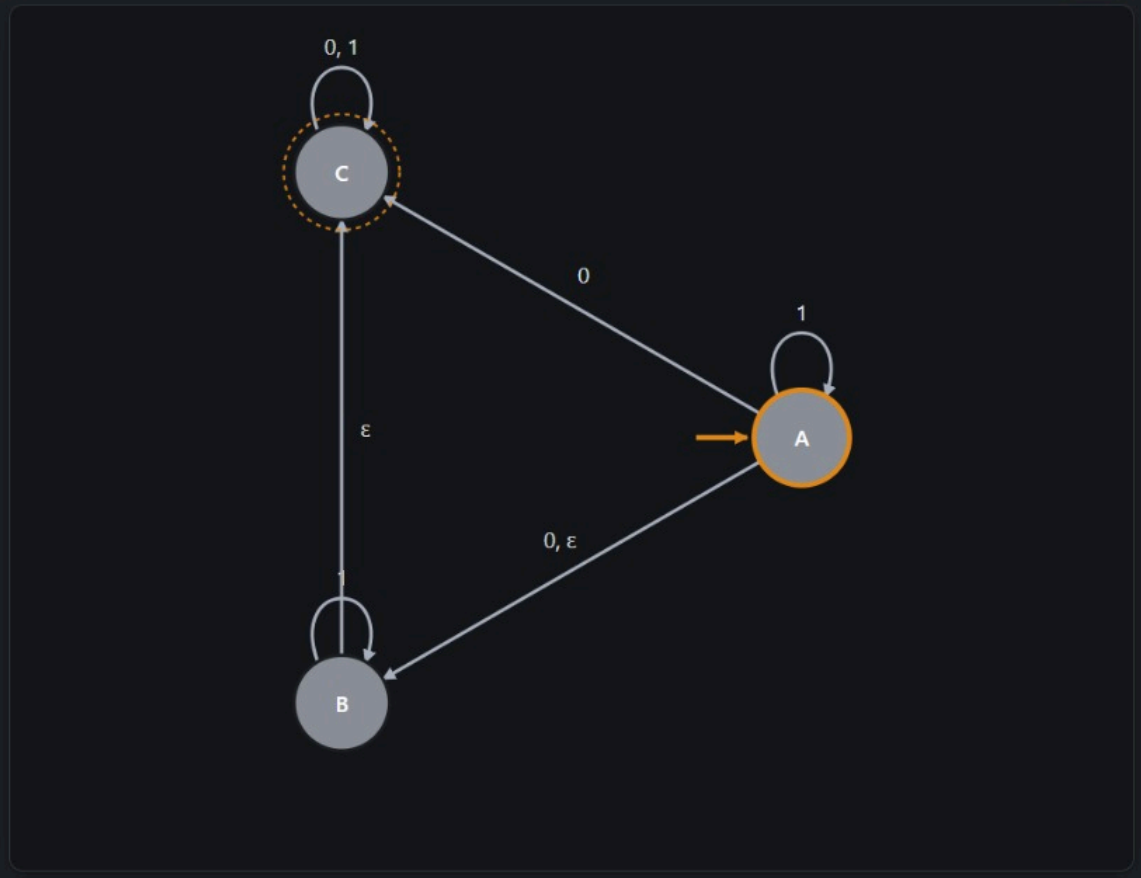
TO
C

Add transition

A —0→ B
A —0→ C
A —1→ A
A —ε→ B
B —1→ B
B —ε→ C
C —0→ C
C —1→ C

Delete transition

DFA: binary strings ending in 1 | NFA



State | Accepting | Start | Dead | Unreachable | Active

Properties | Convert

NFA → DFA (subset construction)

Minimize DFA

DFA → NFA

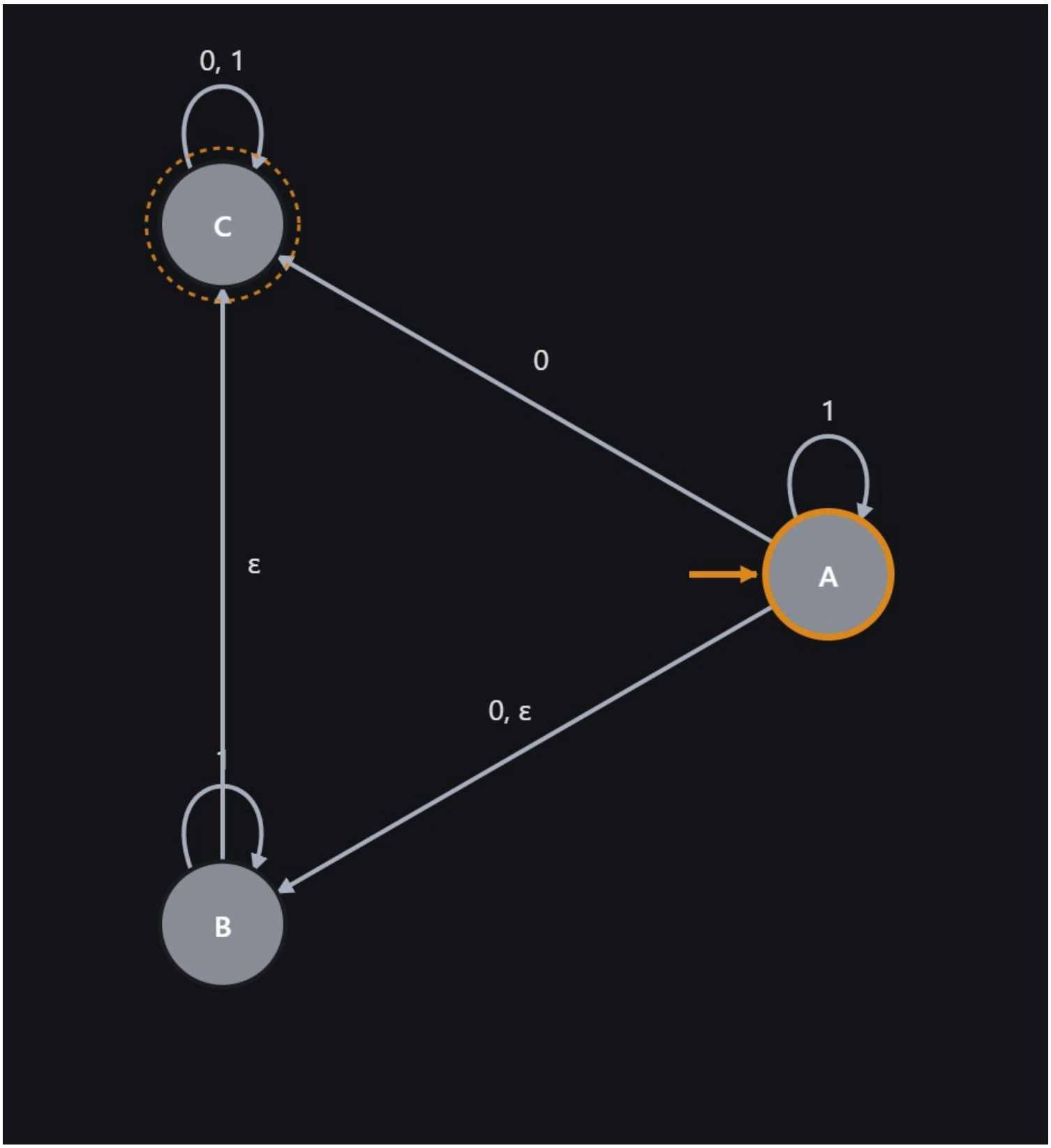
SUBSET CONSTRUCTION TABLE

DFA state	NFA states	0	1
→ {A,B,C}	{A, B, C}	{B,C}	{A,B,C}
{B,C}	{B, C}	{C}	{B,C}
{C}	{C}	{C}	{C}

→ start state · * accepting state · 3 DFA states from 3 NFA states.

Use as current automaton

INPUT STRING e.g. 1011 | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭ | Enter a string and press Simulate.


0, 1
C
0
1
A
ε
0, ε
1
B

AutomataStudio | New | Open | Save | PNG | Report | Examples... | Auto-layout | Fit | + | − | ?

States | Alphabet | Transitions

FROM
q1

SYMBOL
ε (epsilon)

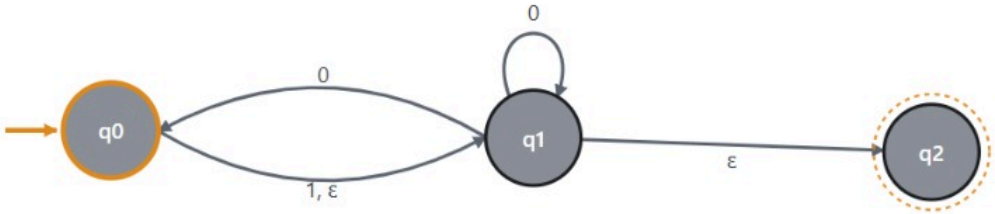
TO
q2

Add transition

q0 —1→ q1
q0 —ε→ q1
q1 —0→ q0
q1 —0→ q1
q1 —ε→ q2

Delete transition

DFA: binary strings ending in 1 | NFA



State · Accepting · Start · Dead · Unreachable · Active

Properties | Convert

NFA → DFA (subset construction)

Minimize DFA

DFA → NFA

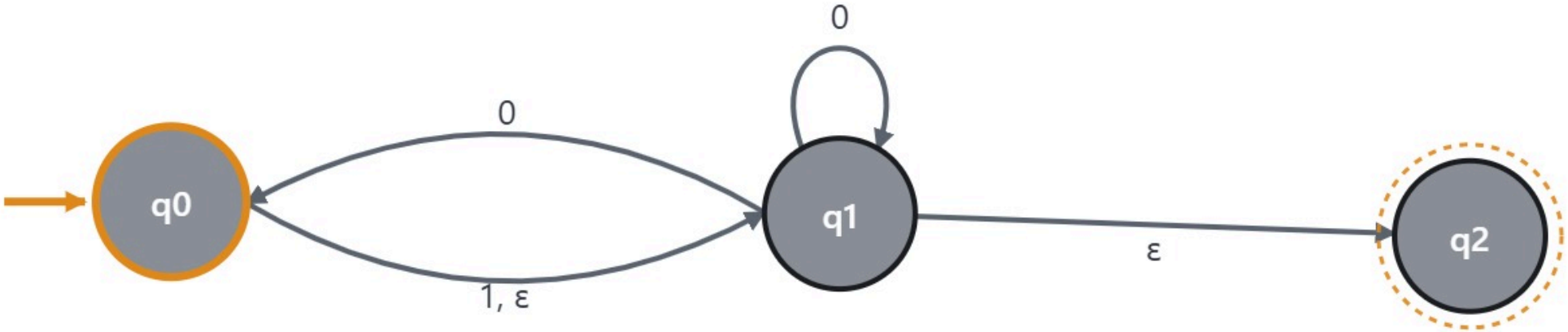
SUBSET CONSTRUCTION TABLE

DFA state	NFA states	0	1
→ {q0,q1,q2}	{q0, q1, q2}	{q0,q1,q2}	{q1,q2}
{q1,q2}	{q1, q2}	{q0,q1,q2}	∅

→ start state · * accepting state · 2 DFA states from 3 NFA states.

Use as current automaton

INPUT STRING e.g. 1011 | Simulate | ⏮ ◀ ▶ Auto ▶ ⏭ | Enter a string and press Simulate.


0
0
q0
q1
q2
1, ε
ε